MIL-A-23871A(SHIPS)
25 August 1967
SUPERSEDING
MIL-A-23871(SHIPS)
11 February 1964
(See 6.3 and 6.4)

MILITARY SPECIFICATION

ANODES, CORROSION PREVENTIVE, LEAD ALLOY ROD SHAPED

1. SCOPE

1.1 This specification covers the requirements for lead alloy anodes for use with impressed current cathodic protection applications.

1.2 Classification. The anodes shall be of the following grades and types, as specified (see 6.2):

Grade A - Lead-silver
Type 1 - Rod, 0.5 inch diameter
Type 2 - Rod, 0.75 inch diameter

Grade B - Lead-antimony-silver
Type 1 - Rod, 0.5 inch diameter
Type 2 - Rod, 0.75 inch diameter

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-T-97 - Tape; Pressure-Sensitive adhesive, Filament Reinforced.

MILITARY
MIL-C-3993 - Copper and Copper Base Alloy Mill Products; Packaging of.

STANDARDS

FEDERAL
FED. TEST METHOD STD. NO. 151-Metals; Test Methods.

FSC 3426

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. The raw materials used shall be such as to produce a lead alloy rod meeting the requirements of this specification.

3.2 Chemical composition. Lead alloy anodes shall conform to the chemical composition requirements shown in Table I.

Table I - Chemical composition.

Element	Grade A	Grade B
Silver, percent, range	1.90 - 2.10	0.95 - 1.05
Antimony, percent, range	-----	5.5 - 6.5
Lead plus Silver, percent minimum	99.5	---
Lead plus Silver plus Antimony, percent minimum	-----	99.5

3.3 Dimension.

3.3.1 Diameter. Lead alloy rods shall be manufactured in a circular cross section, with a minimum diameter of 0.50 inches for type 1 rod, and 0.75 inches for type 2 rod.

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3.3.2 Length. Length of lead alloy rods shall be 72 inches plus or minus 0.5 inch.

3.4 Workmanship. The lead alloy rod shall be of uniform quality and free from bends, kinks, splits, dents, cracks, pits, seams, foreign matter, porosity or other defects that may affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified in the contract or order, a lot shall consist of 5,000 pounds or less of lead alloy anodes from a single melt, or 1,000 pounds or less from the production cast in a single 24-hour period from more than one melt and of the same chemical composition, offered for delivery at one time.

4.2.2 Sampling for fiscal and dimensional examination. From each lot specified in 4.2.1, anodes shall be selected at random in accordance with Inspection Level II and Acceptable Quality Level (AQL) of 2.5 percent of MIL-STD-105.

4.2.3 Sampling for chemical analysis. Samples for chemical analysis shall be selected and prepared in accordance with paragraphs 4.1 and 4.2 of Method 111.1 of Federal Test Method Standard No. 151.

4.3 Visual and dimensional examination. Anodes selection in accordance with 4.2.2 shall be examined for workmanship and dimensions. Where a number of anodes supplied are not in practical conformance with 3.3.1 and 3.3.2 and are equal to or greater than the applicable rejection number of MIL-STD-105, cause for rejection of the entire lot is indicated.

4.4 Chemical analysis. Chemical analysis shall be in accordance with paragraph 5.1 of Method 111.1 of Federal Test Method Standard No. 151. If any sample does not conform to the chemical composition specified in table I, this shall be cause for rejection of the lot represented by the sample.

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4.4.1 Spectrographic. Spectrographic or other suitable techniques may be substituted for wet chemistry analysis.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use.

5.1.1 Packaging. Packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.2).)

5.2.1.1 Packaging and packing, level A and level B. Anodes shall be bundled 8 anodes per bundle. Each bundle shall be bound with three-3/8 x 0.020-inch steel straps, one placed in the center, and one about 1-1/2 feet from each end of the bundle or tape complying with PPP-T-97 comparably used. Wire ties shall not be used. Anodes so bundled shall require no additional wrapping and no preservation, and shall be packed in category 1, 4, or 6 containers in accordance with MIL-C-3993.

5.2.1.2 Marking. In addition to any special marking required, marking for shipment shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Anodes covered by this specification are intended for use with impressed current systems for the cathodic protection of the underwater surfaces of inactive ships. They are designed for suspension in the water alongside the ship's hull in sea water locations and should not be used in fresh water or mud.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Grade and type of anode required (see 1.2).
(c) Packaging, packing or marking requirements other than those required by paragraph 5.1 (see 5.2).

6.3 Supersession data. MIL-A-23871A(SHIPS), grade A includes the requirements of composition I and grade B includes the requirements of composition II.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

6.5 Rods. The term "rods" refers to the general form and dimensions of the material, and does not have any technical significance as to the methods of manufacture.

Preparing activity:
Navy - SH
(Project 3426-N017)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated.. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 61 0506
(NAVSHIPS OVPRT 12-66)

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